

The Diamond First Law: Interval Modular Hamiltonians, the Local Temperature Weight, and the Entanglement-Equilibrium Formulation

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Abstract. The first-law paper established $\delta S = \delta \langle K \rangle$ exactly for coherent excitations on wedges. This note extends the law to causal diamonds — the geometry on which the entanglement-equilibrium derivation of the Einstein equations operates — and then states that derivation as a formulated problem in the model's own objects. The interval modular Hamiltonian carries the Hislop-Longo weight, $K_{(a,b)} = 2\pi \int_a^b v_{ab}(r) T(r) dr$ with $v_{ab}(r) = (r-a)(b-r)/(b-a)$ — the vector field of the Möbius interval flow this series verified machine-exactly in the geometrization paper — reducing to the wedge weight r as $b \rightarrow \infty$, with the local inverse temperature $\beta_{\text{loc}}(r) = 2\pi v_{ab}(r)$ vanishing at the diamond's edges. **Theorem (the diamond first law):** for coherent waves supported in the interval, $S(\omega_F || \omega)|_{(a,b)} = \Delta \langle K_{(a,b)} \rangle = 2\pi^2 \int v_{ab} F'^2 dr$, exactly — same four-step proof as the wedge case. Verified by an independent route forced by the geometry: the interval generator admits no Fourier diagonalization, so the spectral side is computed as the *flow-derivative of the one-particle pairing along the verified Möbius flow*, agreeing with the classical weighted energy at $1.7 \times 10^{-6}, 1.2 \times 10^{-6}, 3.2 \times 10^{-6}$ across three configurations; the wedge limit obeys its exact finite- b deficit law with the predicted coefficient at sub-percent accuracy. Through the assembly, axis-tipped double cones in 3+1 correspond to *uniform* per-generator intervals $(t_1/2, t_2/2)$, so the diamond law holds in the bulk for all diamonds with tips on lines through the apex, Lorentz transporting among them. With wedges and diamonds both carrying the exact law, the entanglement-equilibrium problem is stated precisely: the geometry enters through the weight v — the diamond's conformal Killing data, a metric functional — and the demand that $\delta S = \delta \langle K \rangle$ persist under joint metric and state variations for all small diamonds is the constraint whose solution is the linearized Einstein equation. Formulated, not claimed; the execution is the gravity thread's next step.

1. From wedges to diamonds

The entropic route to gravity runs through causal diamonds: Jacobson's equilibrium argument varies the entanglement of small diamonds, not wedges. The previous paper proved the exact first law on wedges; this note carries it to diamonds, whose modular structure the series already holds twice over — abstractly through Hislop-Longo, and concretely through the geometrization paper's machine-exact verification of the Möbius interval flow as a symmetry of the state.

2. The interval modular Hamiltonian and the local temperature

The modular flow of the interval algebra $\mathcal{A}((a, b))$ in the vacuum is the Möbius flow fixing a and b ; its vector field is $2\pi v_{ab}(r)$ with

$$v_{ab}(r) = \frac{(r-a)(b-r)}{b-a}, \quad K_{(a,b)} = -\ln \Delta_{(a,b)} = 2\pi \int_a^b v_{ab}(r) T(r) dr,$$

$T = \frac{1}{4\pi} :j^2:$ the stress tensor derived in the companion paper. Conventions stated once: Δ^{it} implements the flow at modular time t (the 2π lives inside the flow), run in the Bisognano-Wichmann backwards orientation, so that K is positive on interval excitations. Two readings of the weight. As $b \rightarrow \infty$, $v_{ab}(r) \rightarrow r$: the wedge Hamiltonian is the diamond's limit, and the finite- b correction is exact and rational, $\Delta \langle K_{\text{wedge}} \rangle - \Delta \langle K_{(a,b)} \rangle = 2\pi^2 \int r(r-a) F'^2 dr / (b-a)$. And locally, $\beta_{\text{loc}}(r) = 2\pi v_{ab}(r)$ is the diamond's position-dependent inverse temperature, vanishing at both edges — the thermal reading of confinement to a finite causal region.

3. Theorem: the diamond first law

****Theorem.**** For real $F \in C_c^\infty((a,b))$, with $\ell = 2\pi F'$ the coherent wave of $W(F)$,

$$S(\omega_F \parallel \omega) \big|_{\mathcal{A}((a,b))} = \Delta \langle K_{(a,b)} \rangle = 2\pi \int_a^b v_{ab}(r) \frac{\ell(r)^2}{4\pi} dr = 2\pi^2 \int_a^b v_{ab}(r) F'(r)^2 dr,$$

exactly.

Proof. The four steps of the wedge case transfer verbatim: the unitary identity $S(\omega \circ \text{Ad } U^* \parallel \omega) = \langle U\Omega, (-\ln \Delta) U\Omega \rangle$ with $U = W(F) \in \mathcal{A}((a,b))$; the interval modular structure $-\ln \Delta_{(a,b)} = d\Gamma(K_1)$ with K_1 the one-particle generator of the (backwards) Möbius flow — Hislop-Longo, and the geometrization paper's verified invariance; the Weyl displacement reducing the quadratic generator to its one-particle value on the shift; and the identification of that quadratic form with the classical weighted energy. ■

4. Verification

The interval generator has no Fourier diagonalization, which forces — and thereby strengthens — an independent verification route: the spectral side is computed as the flow-derivative of the one-particle pairing, $\langle KF, K_1 KF \rangle = \mp i \frac{d}{dt} \langle KF, K(F \circ \varphi_{-t}) \rangle \big|_{t=0}$, with the flowed test function composed analytically through the exact Möbius map and the pairing evaluated in Fourier. Against the classical side $2\pi^2 \int v_{ab} F'^2$: relative deviations $1.69 \times 10^{-6}, 1.21 \times 10^{-6}, 3.22 \times 10^{-6}$ for three interval-packet configurations, imaginary parts at quadrature noise. The wedge-limit deficit law was tested separately: the predicted coefficient $2\pi^2 \int r^2 F'^2 = 36.054$ against observed ($b \cdot \text{deficit}$) values 36.09, 36.18 at the well-resolved windows — sub-percent agreement, with the exact rational form of §2 behind it.

5. Diamonds in the bulk, through the assembly

An axis-tipped double cone in 3+1 — tips $(t_1, \vec{0})$ and $(t_2, \vec{0})$ — meets every generator of the assembly's cone in the *same* interval $(t_1/2, t_2/2)$: the bulk diamond's characteristic data footprint is a uniform interval family, and its modular Hamiltonian is the generator sum of the chiral interval Hamiltonians above. The diamond first law therefore holds in the bulk for all axis-tipped diamonds, and Lorentz invariance (the assembly paper's output)

transports it to every diamond whose tips lie on a timelike line through the apex — each such diamond being axis-tipped in its own frame. Diamonds in fully general position relative to a fixed cone give generator-dependent intervals whose joint modular flow does not preserve the cone; that case belongs to the 3+1 Hislop–Longo machinery directly and is cited, not re-derived.

6. Entanglement equilibrium, formulated

What is now proven inside the model: the exact first law $\delta S = \delta \langle K \rangle$ on wedges and on diamonds, on a dense family of excitations, with K 's carrying the geometric weights r and v_{ab} and S the Araki entropy on the series' type III_1 algebras. What the entanglement-equilibrium derivation adds, stated in the model's own terms: the weight v is the diamond's *conformal Killing data* — a functional of the metric — so a geometry perturbation enters the law on the $\langle K \rangle$ side through δv , while a state perturbation enters both sides as above.

Jacobson's postulate is that the vacuum is an equilibrium: for all sufficiently small diamonds, in all frames, the first law persists under *joint* metric and state variations. The content of that demand is a constraint on δg , and its solution is the linearized Einstein equation, with the equation-of-state reading — the metric responding as a thermodynamic variable, never an operator — exactly the state-stratum account of gravity this series adopted in its opening paper. Here that program stops being a slogan: every symbol in the demand now denotes a constructed object of this archive. The execution — the diamond-deformation calculus and the extraction of the constraint — is the gravity thread's next step, and the two standing cautions travel with it: the stress-tensor improvement ambiguity contributes edge terms for waves crossing the boundary (Casini–Grillo–Pontello), and interactions, when they arrive, add energy exchange and the soft-sector dynamics.

7. Scope

Five items. (i) The theorem covers interval-supported waves; boundary-crossing waves are the cited harder case with the improvement-term caution. (ii) General-position diamonds are cited to the 3+1 Hislop–Longo structure. (iii) The entanglement-equilibrium execution — geometry variation, the constraint, the linearized equation — is the next brick of the gravity thread. (iv) The measurement gate (the Fewster–Verch probe on this net) remains the era's Step 2. (v) The covariance audit of the tower's finite- μ remnants remains Step 3, the era's central open risk.

References

- Companion papers: *The First Law on the Net: The Stress Tensor, the Wedge Modular Hamiltonian, and the Relative Entropy of Coherent Excitations*; *Geometrization: The Exponential Correspondence, Thermal Relative Duality, and the Shadow's Home*; *The Interior Net: Local Algebras from the Cone and Microcausality as Output*; *The Assembly: Null Lines Through a Point, the Light-Cone Identification, and Lorentz Symmetry as Output*.
- Hislop, P. D. & Longo, R., on the modular structure of local algebras of the free massless field, *Comm. Math. Phys.* 84 (1982).

- Jacobson, T., "Thermodynamics of spacetime: the Einstein equation of state," *Phys. Rev. Lett.* 75 (1995), 1260.
 - Jacobson, T., "Entanglement equilibrium and the Einstein equation," *Phys. Rev. Lett.* 116 (2016).
 - Longo, R., "Entropy of coherent excitations," *Lett. Math. Phys.* 109 (2019), 2587–2600.
 - Casini, H., Grillo, S. & Pontello, D., "Relative entropy for coherent states from Araki formula," *Phys. Rev. D* 99 (2019), 125020.
 - Bisognano, J. J. & Wichmann, E. H., "On the duality condition for a Hermitian scalar field," *J. Math. Phys.* 16 (1975), 985–1007.
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